Synopsis

on

Personalized Course Recommendation System

to be developed to fulfil the requirements for

Major Project

Submitted to

Department of Computer Applications

CGC University, Mohali, Punjab

under the supervision of

Mr. Mohit Ruhil

Professor

Submitted by

Shweta	Shivani	Simarpreet Kaur
Roll no.2335225	Roll No.2335213	Roll no.2335227
Section GH	Section GH	Section GH

BCA 6th Sem

(Batch 2023-26)

TABLE OF CONTENTS

6. References

1. **Abstract**

The Personalized Course Recommendation System is a web-based application designed to help students discover educational content based on their specific interests. By utilizing **unsupervised machine learning (Clustering Algorithms)**, the platform categorizes courses into similar groups and matches them with user preferences with available courses. Providing a streamlined and efficient alternative to manual searching. This project focuses on accessibility and simplicity, ensuring that users with basic technical knowledge can find relevant learning paths quickly.

1. **Introduction**

This chapter gives an overview about the aim, objectives, background and operational environment of your project.

2.1 Project Aim and Objectives

The primary aim is to build an interactive web platform that simplifies the process of finding educational courses through automated matching..

Key objectives:

- Enable user interaction through a clean, HTML-based web interface.
- Develop a python-based **clustering engine** to group courses by shared attributes (e.g., difficulty, duration, and keywords).
- Provide instant, personalized recommendation based on user-selected interests.
- Reduce the time spent by students browsing through irrelevant academic content.

2.2 Technology to be used

- Frontend: HTML, CSS, JavaScript.
- Backend: Python(Flask framework) **Scikit-learn** (for the clustering algorithm), **Pandas** (for data manipulation), and **NumPy**..
- Data Handling: **CSV/JSON datasets** handled via Pandas for more robust data clustering.

2.3Hardware and Software Requirement

Processor	i5 processor
Operating system	Windows 11
Memory	8GB
Hard disk space Software version	512GB 1.105

3. System Analysis

3.1 SRS(Software Requirement Specification)

- Users: Students/Learners.
- Functional Requirements: The system accept user interest inputs, process them against a course catalog, and display matching results.

- Non-functional: Fast response times, user friendly UI, and platform-independent accessibility via web browsers .

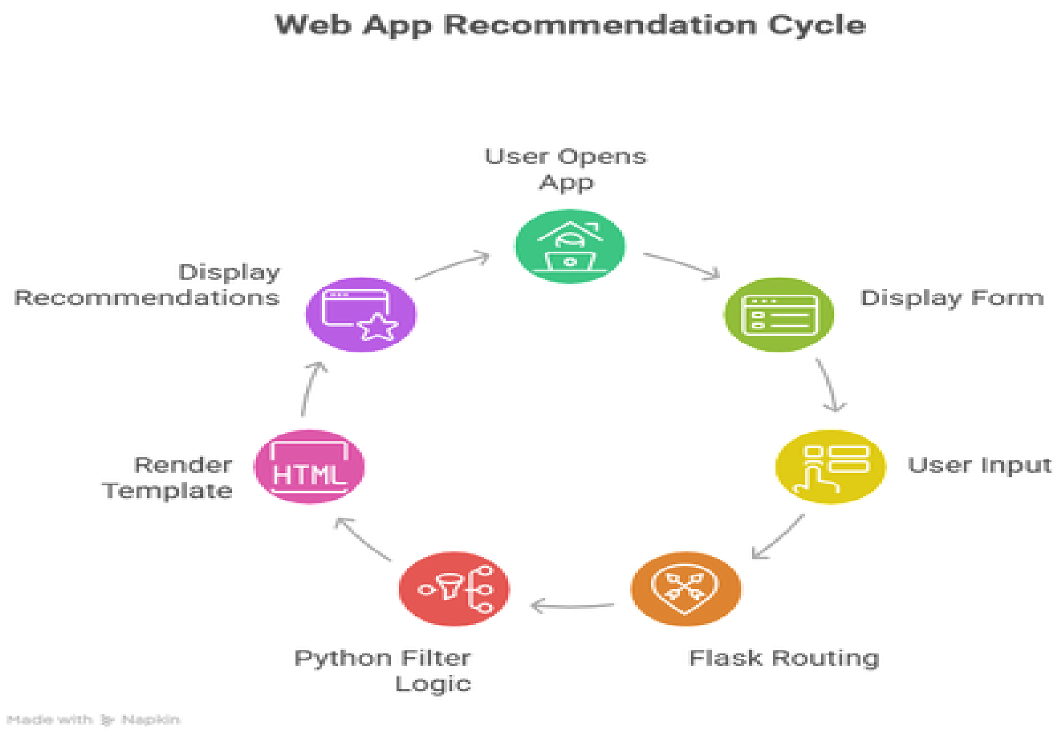
3.2 Existing Vs Proposed System

Aspect	Existing system	Proposed system
Search method	Manual browsing of long lists	**Algorithmic clustering** for pattern-based discovery
Speed	Slow and tedious	Instant matching and display
User effort	High(must read every title)	Low(select interest from dropdown
Accuracy	Prone to human oversight	Logical consistent and pattern-driven results.

3.3 Software Tools to Be Used

- Visual Studio Code/Sublime Text (coding).
- Manual(F12), Flask client, Pytest (testing).
- Web browsers (UI/UX testing).
- Draw.io for DFD/ER/Flowchart diagrams.
- Matplotlib or Seaborn.

4. Data Flow Diagram



5.Timeline

Synopsis Preparation	**Feb 1-5, 2026**
Requirement analysis and design	**Feb 6-20, 2026**
Development	**Feb 21- Mar 20, 2026**
Testing & Debugging	**Mar 20-30, 2026**
Final Submission	**Apr 15, 2026**

6.References

1. “Flask mega-tutorial: Building a web app with a python,” Miguel Grinberg,2024.
2. “Build a simple web application using python and flask,”GeeksforGeeks,2024.
3. “Python for beginners: working with list and dictionaries, “Real Python,2023.
4. “Creating dynamic web pages with HTML and Flask templates,”W3Schools,2025.
5. “A beginner’s guide to rule-based Recommendation Systems, “Towards data science(Archive),2022.